

Technology: There is little to differentiate between an LCD and a DLP projector. Most people are likely to opt for a DLP solution because it is normally the smaller of the two and the images quality, especially where movies are involved, is slightly better.

Resolution: Resolution determines the sharpness and clarity of an image. The most common resolution is SVGA or 800x600 pixels. This is sufficient for most users and SVGA projectors are the cheapest available in the market.

Define your task here—if you are looking at meetings with nothing but presentation slides, SVGA is good enough. But if you are part of a sales team and find yourself pitching spreadsheets full of projected growth and earnings to your immediate superiors, you might want to look for an XGA or a 1,024x768 pixel projector. The larger number of pixels that an XGA solution affords will bring greater clarity to the small text so prevalent in a spreadsheet. You can take this forward: the finer the detailing on your presentation (maybe a CAD drawing?) the higher the resolution required and the greater the price you will pay. You thus have SXGA (1,280x1,024) and UXGA (1,600x1,200) solutions, the latter being a rare breed due to their high asking price.

One final point to consider—a projector has a “native” resolution; the native resolution for an SVGA projector will be 800x600, for an SXGA unit 1,280x1024 and so on. Try and match the native resolution of your projector to the screen resolution of either your laptop or the PC from which you are sourcing the presentation. If the projector’s native resolution is different from your computer’s screen resolution, the output



HP vp6110

will appear fuzzy due to “scaling”.

Brightness: This directly affects the size of the audience your projector can address. The brighter the projector the better and you should always go for the brightest unit you can afford. Define the nature of your presentation first. If your business showcases more videos or if you are using the projector at home, brightness loses importance since these tasks are usually done in a dark room—and visibility is therefore not a problem. If, however, you are presenting to a group of note-takers or are teaching a work-group, your room will need some sort of illumination, and so your projector’s brightness comes to the fore. You can alleviate the need for bright units (since the associated cost increases with brightness) by employing reflecting screens for your projecting. Note that a typical wall is generally the worst place to project upon.

Finally, the size of your audience will also play a part. The larger your audience, the larger and brighter the projected image will need to be.

Brightness is measured in ANSI lumen or just lumen. An entry-level projector carries 1,000 lumen or less; as this rating increases, so does the price. A 3,000 lumen projector will classify as a high-end unit but a 12,000 lumen unit is not

Brightness	Usage
1,000 lumen or less	Make your presentation in a dark or dimly lit room
1,000 - 2,000 lumen	Suited for small classrooms or small presentation room
2,000 - 3,000 lumen	Suited for large rooms
3,000 lumen or more	Suited for large venues such as auditoriums and board rooms

How We Tested

Test Bed

The test bed for projectors consisted of an ACI Impression M11 laptop with a P IV 2.4 GHz processor running Windows XP Professional SP1. It had 512 MB of DDR SDRAM, and an ATI 9700 graphic processor. All the projectors were tested with Windows Display Properties set to 32-bit colour. The projector’s native resolution as specified by the manufacturer was set before the tests were run.

Preparations

Before running the tests, parameters such as ambient light, distance of the projector from the screen, and the graphic chips used in the test bed were kept exactly the same for all projectors under review. Testing was done under controlled lighting with the distance from the screen set for optimal results, as specified in the users manual by the respective manufacturers. The screen we used measured 72x90 inches. The projector’s distance from the screen was then adjusted for optimal results.

Features

Here, we noted feature sets such as

brightness, contrast ratio, aspect ratio, optical zoom, manual focus, presence of various interfaces such as DVI, D-SUB, and so on and dimensions and weight. We also logged special features such as remotes that doubled as a presentation tool.

Presentation Test

In this test, we ran a PowerPoint presentation, which included a movie clip, regular text on a white background, black text on a coloured background, coloured text on a coloured background, and a graph. This test was run under the ‘Presentation’ mode and rest of the settings were left to factory defaults. This was to see how well the projector performed when used for presentations.

Animation Test

In this test, we ran a Flash file with animation and a clickable button. This test was to evaluate slow moving graphics, and also to check colour reproduction.

Movie Test

A DivX movie in 16:9 aspect ratio was run

from the laptop to evaluate the ‘Theatre’ mode of the projector. The projector was set to an aspect ratio of 16:9, if the particular feature was available; otherwise it was left at native mode.

DisplayMate Video Edition

DisplayMate Video Edition was used to test the image quality of the projectors. The tests are designed to bring out problems like digital noise, moiré patterns, streaking, ghosting and colour accuracy deviations. Twenty-two test screens, which included focus matrix, colour purity and level shift were used to evaluate image quality.

How we awarded

The score noted from features, performance and from the price index was given importance as per as the specified category. An overall score out of 100 was then calculated. The product that scored the highest in the overall score was adjudged the winner of the Digit Best Buy Gold Award for its category. The runner-up received the Digit Best Buy Silver Award.